Epidemiology
Week 4
Routinely Collected Data

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1 Definition and Examples

Routinely collected data is data collected regularly for purposes other than research. It is typically collected by government bodies or public health institutions and follows a strictly defined protocol or template. Routine data usually focuses on economic status, infections / chronic disease, births, mortality, or socio demographic status. Examples of routinely collected data: Census, birth registry, race, educational attainment.

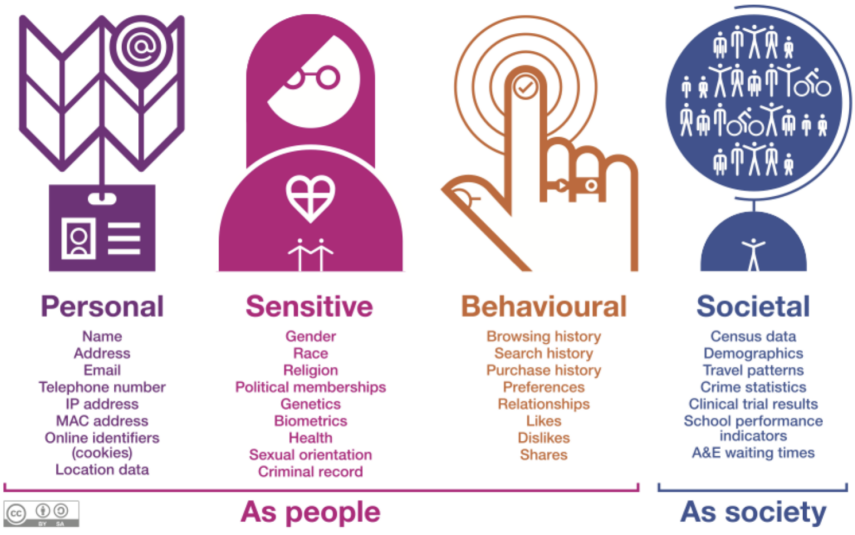


Figure 1: Routinely collected data examples

Sources on some routine data

Demographic data

- Census
- Demographic surveillance
- Community Health Index (Chi) register
- Survey

Circumstantial data

- Education
- Employment
- Housing
- Environmental

National reference data

- Deprivation index
- Health Poverty index
- GINI coefficient

Health event data

- Birth and death registration
- Disease registries
- Hospital admissions
- Global Burden of Disease study

Census

Census is a procedure of systematically enumerating, acquiring, and recording information about the members of a given population, and is usually done every 10 years. It is compulsory for everyone and the grain of the data is both at an individual and household. In some countries instead of census, a population register is used.

2 Use and Challenges

Data evaluation of routine data is evaluated with the acronym CART.

- C: Completeness
- A: Accuracy
- R: Representativeness
- T: Timeliness

While analysis is evaluated with the acronym PACO.

- P: Precision -> How big is the database?
- A: Accessibility -> How is the data accessed?
- C: Confidentiality -> Can individuals be identified?
- O: Original Purpose -> Why was the data collected?

Routine data can be used for

- **Summary statistics**: Measuring prevalence or incidence
- **Trend analysis**: Time series of disease rates
- **Spatial analysis**: Mapping disease by region
- **Exposure group analysis**: Comparing outcomes by risk factors
- **Hypothesis testing**: Observational studies using linked data

Disease occurrence is measured by

- **Numerator**: Number of cases
- **Denominator**: Population at risk
- **Mortality measures**:
  - Age-specific mortality, typically expressed per 1,000 or 100,000 people
  - Cause-specific mortality, typically expressed per 100,000 people
  - Crude mortality, typically expressed per 100,000 people
  - Infant mortality, typically expressed per 1,000 live births
  - Maternal mortality, typically expressed per 100,000 live births

Population pyramids

Population pyramid is a graphical representation of age and sex distribution and is used to identify demographic trends, aging populations, and birth rates. The types are shown below

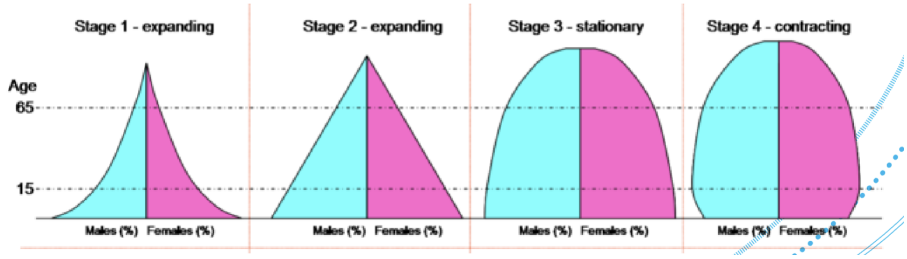


Figure 2: Population pyramid types

2.1 Sources of bias in routine data

Selection bias

- When data collection systematically excludes a population group
- E.g. illegal immigrants, socially deprived groups

Information bias

- When self reported data systematically under or over-reports, e.g. Smoking status
- Can be caused by non-standardisation of variables, e.g. Different scales to measure deprivation

2.2 Linkage studies

Linkage study is the act of combining multiple datasets, such as routine data, using unique identifiers (like NHS number, names, DOB) to track individuals across different systems.

- To **follow individuals over time** across different health services
- To **study associations** between exposures and outcomes
- To **enhance data completeness** and reduce bias
- To **enable longitudinal research** without needing new data collection

Advantage	**Disadvantage**
Large sample size / whole population	No researcher input, linited variables
Baseline data, useful for initial assessment	Data quality being incomplete or inaccurate
Long term follow-up	Availability timing
Reduced risk of recall bias	Occasionally subject to manipulation or political influence
Readily available and cost-effective	Subject to privacy and consent
Richer and possibly more accurate	Complex (in analysis and approvals)

Table 1: Pros and cons of linkage study

3 Epidemics

- Endemic: Commonly present at baseline level in given geographical area or population
- Epidemic:
  - Excess cases of a disease compared with expected number
  - Generalised, cases distributed across wider area / time period
- Outbreaks:
  - Excess cases of a disease compared with expected number
  - Limited to a localised increase in disease (often infectious) incidence, e.g. town, village, institution
- Pandemic: Affects large numbers of people and crosses international boundaries

3.1 Types of epidemics

Classified by source of infection and mode of transmission, epidemics can be divided into three

- Common source outbreaks
- Propagative or progressive epidemics
- Mixed epidemics

Common source outbreaks

Group of individuals exposed to a common source pathogen, often at the same time or within a short period of time. **Point source outbreak** is a common source outbreak where exposure period is short, and all cases who fall ill have the same **incubation period**. Common source outbreaks have **tight temporal clustering** and the epidemic curve follows a sharp upslope with gentle downslope. An example is foodborne diarrhoeal illnesses.

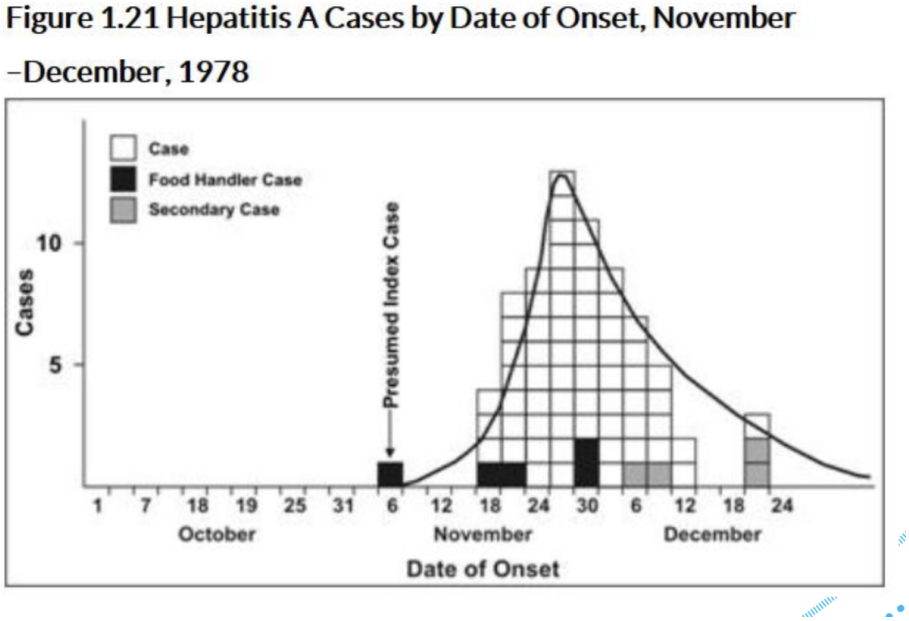


Figure 3: Example of common source outbreak epidemic curve

Propagative or progressive

Propagative epidemic is when the infection spreads from person to person directly or indirectly via vectors (e.g. mosquitoes, food, water). Cases usually occur over more than one incubation period. The epidemic curve follows a gentle upslope and steeper tail with time period between peaks consistent with the disease's incubation period.

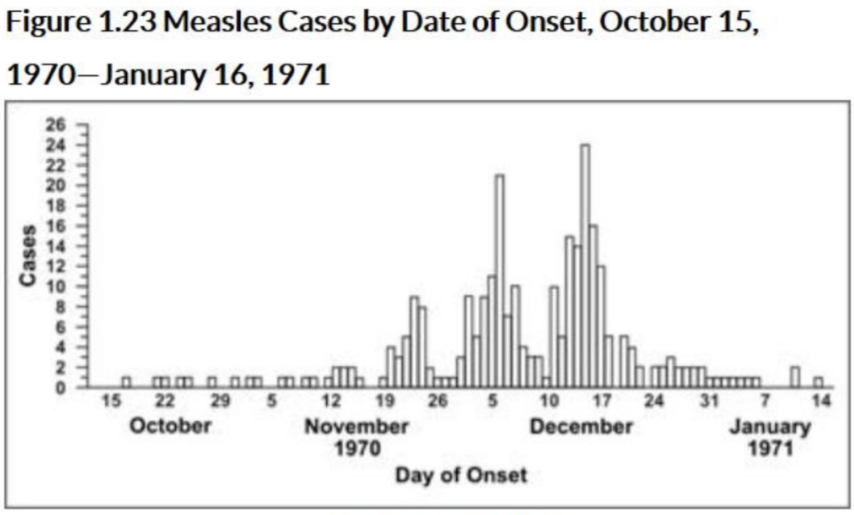


Figure 4: Example of propagative epidemic curve

Mixed epidemics

Has both characteristics of common source outbreaks and propagated epidemics. It often starts from a point source (common source) infectious agent followed with subsequence propagative spread. E.g. Shigella, typhoid.

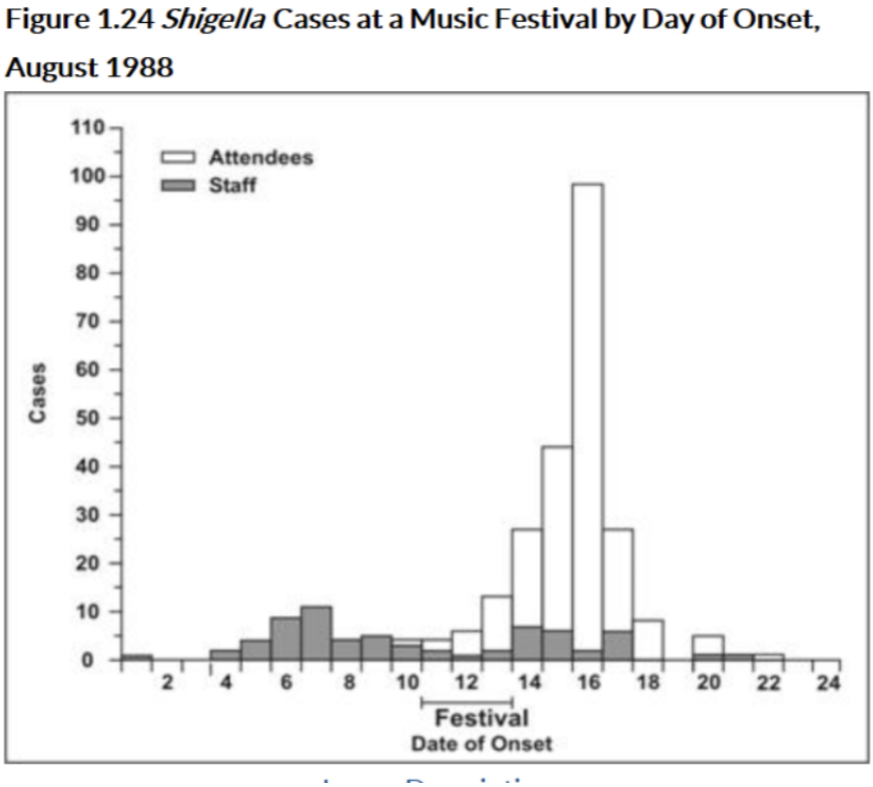


Figure 5: Example of mixed epidemic curve

4 Epidemic Investigation

Investigation in epidemics follow these three process:

1. Identification
2. Hypothesis generation and testing
3. Confirmation and control efforts

Starting with identification, the body or person in charge of identifying is usually clinicians, local and national health agency for disease surveillance (e.g. Health Protection Scotland, Public Health England, CDC), or even the public.

4.1 Identification

- Determine existence of epidemic
- Confirm diagnosis
- Define and count cases
- Orient data in terms of time, place, and person
- Determine who is at risk

4.2 Hypothesis generation and testing

- Develop and test hypothesis relating to the cause of epidemic
- Compare hypothesis with current facts

Studies can be done by running a case-control study or a cohort study. However, when time is tight, a cohort study might not be possible.

Case-Control Study	**Cohort Study**
Identifies people with disease and compare against people without disease	Identifies people **exposed** to a particular factor and compares against people not exposed to the factor
Cases and controls compared in terms of frequency of one or more past exposures	Measures and compares incidence of disease
If cases have **higher odds** of exposure to factor than controls (odds ratio), this would suggest association	Magnitude of association expressed as risk ratio
Good choice if source population is **large** and ill-defined, and disease **uncommon**	Good choice if small, relatively well-defined source population, and disease fairly **common**

Table 2: Case-control vs cohort study

4.3 Confirmation

Once the study is done, an action can be done which is usually in the form of control and prevention activity which may:

- Control / modify host response to exposure, e.g. Vaccination
- Interrupt transmission
- Control source of pathogen, e.g. Isolate and treaet infected people, remove source of contamination

Furthermore, a clear plan and execution of a systematic study with a detailed report.

Biases and challenges in epidemic investigation

- **Sampling bias**; People may not wish to cooperate with the study, loss at follow-up, difficulty in contacting people
- **Measurement bias**; Lab confirmation may not be possible, misclassification of cases
- **Small sample size**; Lack of sample size affects statistical power
- **Publicity**; Can encourage participation and uptake of control measure but may also lead to preconceptions about causes
- **Lack of public health infrastructure**

5 Cluster

Cluster is an aggregation of *'relatively uncommon events / diseases in space and/or time in amounts that are believed or perceived ot be greater than could be expected by chance'*. This can include infectious or non-communicable conditions such as cancer, spontaneous abortions. It often has unknown or multiple aetiological factors, low exposure level over number of years. It may come from environmental or occupational causes. Examples of cluster investigations:

- Rare diseases: Pneumocystis jirovecii pneumonia and HIV
- Birth defects: Thalidomide and limb defects
- Cancers: Asbestos and mesothelioma, hydrocarbons (soot) and scrotal cancer

5.1 Initial investigation

The steps taken for the initial investigation are

1. Initial contact and response
   - Is it worth following up?
   - Does it meet the definition of a cluster?
   - Is there a biological possibility that the reported disease could share a common aetiology
2. Assessment
   - Case ascertainment and case definition refinement
   - Define population at risk
   - Calculate standardised incident ratio $\frac{\text{No of expected cases}}{\text{No of observed cases}}$
3. Determine feasibility of conducting an epidemiological study
4. Conduct the study

5.2 Challenges of cluster investigation

- Rare health events
- Lack of population base for rate calculation
- Weak association and multiple risk factors
- Long induction periods
- Multiple comparisons
- Low level, long term, heterogeneous exposures
- Intense publicity

5.3 Role of public health agencies

- Maintain objectivity
- Develop effective communciation strategies and strong leadership
- Conduct a robust scientific investigation where appropriate
- Sensitive to public concerns
- Respond to community suspicions and demands
- Recognise the role of the media
- Aware of potential adverse social and economic ramifications
- Aware of potential legal ramifications of the investigation